

Figure 1: Radio waves in the electromagnetic spectrum

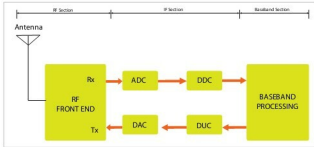


Figure 2: Block diagram of a conventional hardware based radio

machine learning capabilities. Ivan Seskar points out in his hands-on tutorial (Reference 4) that, "Cognitive Radio is an SDR that is frequency-agile, fully reconfigurable and able to sense its spectrum surroundings; it knows policies, rules and regulations, and is flexible enough to reconfigure itself to different air interfaces and/or protocols."

Let's now discuss GNU Radio, as it is one of the most widely accepted tools all over the world.

GNU Radio

The GNU Radio project was conceptualised by Eric Blossom. It is an open source toolkit, which is made of several signal processing and communication blocks written in C++ and mapped onto Python using a simplified wrapper and interface generator (SWIG). Figure 4 shows the typical architecture of GNU Radio. GNU Radio Companion (GRC) is a graphical user interface, which makes dealing with this tool a little easier. It automatically generates source code (in Python) as per the connections between various blocks. Appropriate connection of various signal processing blocks in a signal flow graph creates software radio. It enables users to generate and process waveforms in software with a rich interface.

Before dwelling into the domain of GNU Radio, it is expected that one should have sufficient knowledge of C++/Python, communication systems, digital signal processing and analogue digital hardware (Reference 3).

Installation guide

GNU Radio is available for Linux, Windows and Mac OS. However, it works best in a Linux environment. The installation process on Ubuntu Linux is illustrated in this section. There are three ways to install this toolkit. Because this is not a standalone application and requires many

Tier	Type	Reconfigurability level
0	Hardware Radio (HW)	A digital hardware radio that cannot be altered; reconfiguration through component exchange
i	Software controlled Radio (SCR)	Reconfigurations through control functions in software; limited to predefined set of configuration
ii	Software Defined Radio (SDR)	Software control and reconfigurability of a variety of modulation techniques (waveforms), bandwidth, signal detection, security etc.; frequency constrained
iii	Ideal Software Radio (ISR)	Analogue conversion takes place at antenna, speaker and microphone, everything else is done in software
iv	Ultimate Software Radio (USR)	Understands all traffic and control information and supports (most) applications and radio air interfaces

Figure 3: Software radio: A tiered approach (source: <http://www.sdrforum.org/>)

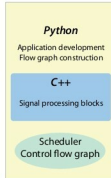


Figure 4: GNU Radio architecture

dependencies to work, one needs to make sure that all the necessary packages are also installed in advance.

Installation through repositories:

This is the beginners' first choice to install the gnuradio package from the Linux distribution's standard repositories. Because GNU Radio is being developed rapidly, one must use the latest binaries to avoid any version related issues at later stages. The following commands install the basic components of GNU Radio and GNU Radio Companion, respectively.

```
$ apt-get install gnuradio
$ apt-get install gnuradio-companion
```

If you need a bootable DVD with GNU Radio preinstalled, follow <https://gnuradio.org/redmine/projects/gnuradio/wiki/GNURadioLiveDVD> to try out GNU Radio without installing it on any machine.

Installation from source: Marcus Leech (Reference 5) has contributed a very useful script (build-gnuradio), which can be used to install different versions of GNU Radio without much effort. For more information, users are advised to refer to <http://www.sbrac.org/files/build-gnuradio>.

Installing manually from source: If you have the experience in Linux to install software packages manually from source, follow these steps:

1. Install the prerequisites.
2. Get the GNU Radio source code by downloading it as a tarball.

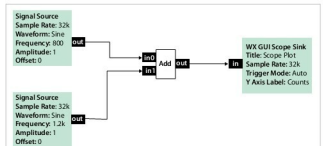


Figure 5: Flow graph for dial tone example in GNU Radio Companion